

PAPER_557: BSFG Symmetry Group and Isometry Analysis — $SO(3) \times U(1)^{23}$ and the DVP 13+13 Partition

Unified Quantum Field Framework (UQFF)

Daniel T. Murphy • UQFF Research Consortium • 2026

Source: PAPER_557_BSFG_Symmetry_Group_Isometry_Analysis.md

PAPER_557: BSFG Symmetry Group and Isometry Analysis — $SO(3) \times U(1)^{23}$ and the DVP 13+13 Partition

Author: Daniel T. Murphy — Star Magic / UQFF Framework **Session:** 148 | **Source:** CP4 #149 (metric) + DVP/VDS number systems **CP4 Class:** [BSFGSymmetryGroupIsometryAnalysisCalculator](#) (#152) **Date:** 2026-03-27

§1 Abstract

The symmetry group of a geometry determines its conservation laws and the redundancies in its coordinate description. This paper derives the complete isometry group of the BSFG manifold $\mathcal{M}^{26}$ by solving the Killing equation $\nabla_{(\mu}\xi_{\nu)} = 0$. The 4D sector has **4 Killing vectors** (time translation + 3 rotations), giving isometry group $SO(3) \times \mathbb{R}_t$. With the 22 compactified dimensions, the full group gains an additional $U(1)^{22}$ factor:

$$G_{\text{BSFG}} = SO(3) \times U(1)^{23} \quad \text{(26 generators total)}$$

A remarkable structural coincidence: 26 generators = 13 stable + 13 destructive modes — exactly the DVP 13+13 partition identified in the arithmetic geometry of BSFG. The VDS eigenvalue triplet $(P/3, P/3, 2P/3)$ provides the $SO(3)$ Casimir invariant, while the $\mathbb{Z}$ temporal symmetry $\cos(\pi(t_{n+1})) = -\cos(\pi t_n)$ constitutes a discrete isometry.

§2 The Killing Equation

A vector field ξ^μ is a Killing vector of $A_\mu{}^\nu$ iff:

$$\nabla_\mu \xi_\nu = \frac{1}{2} \left(\partial_\mu \xi_\nu + \partial_\nu \xi_\mu \right) - \Gamma^\alpha_{\mu\nu} \xi_\alpha = 0$$

On the diagonal metric $A_\mu{}^\nu(r)$, this reduces to:

$$\partial_\mu (A_\nu{}^\nu \xi^\nu) = \Gamma^\alpha_{\mu\nu} A_\alpha{}^\alpha \xi^\nu \quad \text{(no sum)}$$

§3 Killing Analysis — 4D Sector

Test 1 — Time translation: $\xi^\mu = (1, 0, 0, 0)$.

$$\begin{aligned} \nabla_{(0)} \xi_{(0)} &= \partial_0 A_{00}/2 = 0 \text{ since } A_{00} = 1 + \epsilon \text{ depends only on } r, \text{ not } t. \\ \nabla_{(r)} \xi_{(0)} &= \partial_r (A_{00} \xi^0)/2 - \Gamma^0_{r0} A_{00} \xi^0/2 = \epsilon'/2 - \epsilon'/2 = 0. \quad \checkmark \end{aligned}$$

Time translation is a Killing vector.

Test 2 — Rotations: $\xi^\mu = L_{ij}$ (angular momentum generators).

$A_\mu{}^\nu(r)$ is spherically symmetric (depends only on $|r|$, not on angular coordinates θ, ϕ). All three angular Killing vectors ∂_ϕ , ∂_θ , and the third rotation generator are Killing vectors of any spherically symmetric metric. $\checkmark$

Three rotational Killing vectors; isometry group includes $SO(3)$.

Test 3 — Radial translation: $\xi^\mu = (0, 1, 0, 0)$.

$$\nabla_{(r)} \xi_{(r)} = \partial_r A_{rr}/2 = \epsilon'/2 \neq 0 \text{ at any finite } r.$$

Radial translation is NOT a Killing vector — broken by the Aether gradient $\epsilon'(r)$.

Summary (4D sector): 4 Killing vectors = $\{\text{time translation, } L_x, L_y, L_z\}$, isometry group $\cong \mathbb{R}_t \times SO(3)$.

§4 $\mathbb{Z}_2$ Discrete Symmetry

From PAPER_417 (CP4 #67), the temporal modulation $\cos(\pi t_n)$ satisfies:

$$\cos(\pi(t_n + 1)) = -\cos(\pi t_n)$$

Under $t_n \rightarrow t_n + 1$: $\epsilon \rightarrow -\epsilon$. This is a **discrete $\mathbb{Z}_2$ symmetry** of the action — the metric $A_\mu{}^\nu$ transforms to $A_\mu{}^\nu - 2\epsilon \delta_\mu{}^\nu$. While not a

continuous isometry, it is a discrete symmetry of the BSFG theory corresponding to temporal field reversal (the negative time branch of pi-cycles).

§5 The Full 26D Isometry Group

Each of the 22 compactified dimensions θ_i (for $i=5, \dots, 26$) carries a $U(1)$ rotational symmetry ∂_{θ_i} , since the metric coefficient $L_i^2(r)$ is independent of θ_i itself. Adding these to the 4D sector:

$$G_{\text{BSFG}} = \underbrace{SO(3)}_{\text{4D sector, 4 generators}} \times \underbrace{U(1)^{22}}_{\text{22 compactified, 22 generators}}$$

At macroscopic scales where $L_i \rightarrow 0$, the $U(1)^{22}$ sector decouples. But as a formal group structure, the **total number of continuous generators is:**

$$\dim G_{\text{BSFG}} = 3 + 1 + 22 = 26$$

§6 The DVP 13+13 Partition of 26 Generators

The DVP (Dimensional Value Pair) number system (PAPER_540–548) identifies a natural partition of any 26-dimensional structure into **13 stable + 13 destructive** modes. This paper identifies the geometric realization:

DVP Partition	Geometric Realization	Generators
13 stable modes	$SO(3) \times \mathbb{R}_t \times U(1)^9$	$3 + 1 + 9 = 13$
13 destructive modes	$U(1)^{13}$ (remaining compactified dims)	13
Total	G_{BSFG}	26

The "stable" generators correspond to symmetries that preserve the buoyancy condition $F_U^{\text{bi}} \geq 0$ (positive Aether pressure); the "destructive" generators correspond to symmetries that can reverse the sign of F_U^{bi} (gravitational collapse modes). The DVP 13+13 partition is thus a **dynamical stability classification of the isometry group**.

§7 VDS Eigenvalues and the $SO(3)$ Casimir

The VDS (Vacuum Density Spectrum) number system produces eigenvalue triplet $(P/3, P/3, 2P/3)$. Under the $SO(3)$ action, the three eigenvalues transform as components of a pseudo-vector in $\mathfrak{so}(3)^* \cong \mathbb{R}^3$. The $SO(3)$ Casimir invariant is:

$$C_{\rm SO(3)} = e_1^2 + e_2^2 + e_3^2 = \left(\frac{P}{3}\right)^2 + \left(\frac{P}{3}\right)^2 + \left(\frac{2P}{3}\right)^2 = \frac{6P^2}{9} = \frac{2P^2}{3}$$

The $2P/3$ eigenvalue is the **unique orbit** under $SO(3)$ — it has a distinct magnitude from the degenerate pair $(P/3, P/3)$. This eigenvalue structure is preserved by the $SO(3)$ isometry, confirming VDS is a representation of the $SO(3)$ sector of $G_{\rm BSFG}$.

§8 Conservation Laws from Noether's Theorem

By Noether's theorem, each Killing vector generates a conserved charge:

Killing Vector	Conservation Law
∂_t	Energy $E = A_{00}(\dot{t}) = \text{const}$ along geodesics
$L_z = \partial_\phi$	Angular momentum $L = A_{\phi\phi}(\dot{\phi}) = \text{const}$
L_x, L_y	Two more angular momentum components
∂_{θ_i}	Kaluza-Klein charge $q_i = L_i^2(\dot{\theta}_i) = \text{const}$

The broken radial symmetry implies **no conservation of radial momentum** in BSFG — instead, the radial equation of motion includes the Aether fifth force Δg_r from PAPER_555.